

Supplementary material for “Fusing short-pulse voltage waveforms with handcrafted features for rapid state-of-health estimation of lithium iron phosphate batteries: a controlled comparison of encoder pairings”

[First Author]^a, [Second Author]^a, [Corresponding Author]^{a,*}

^a [Department, Institution, City Postcode, Country]

Contents

Supplementary Note S1. Handcrafted feature definitions

Supplementary Note S2. PatchTST encoder equations

Supplementary Note S3. Hyperparameter grids of the conventional regressors

Supplementary Note S4. Statistical test families

Supplementary Note S5. Waveform increment over the GBDT backbone

Supplementary Note S6. Re-implementation used for the generalisation, robustness and deployment experiments

Supplementary Algorithm S1; Supplementary Tables S1–S7; Supplementary Figs. S1–S5

Supplementary Note S1. Handcrafted feature definitions

This note defines the 143 waveform-derived features of Table 2; the seven context quantities are given in Eq. (3) of the main text. The segments follow the publisher's pulse timing and are defined, together with the first and second differences, as

$$S^{(1)} = (V_1, \dots, V_{30}), S^{(2)} = (V_{31}, \dots, V_{40}), S^{(3)} = (V_{41}, \dots, V_{100}), d_t^{(1)} = V_{t+1} - V_t, d_t^{(2)} = d_{t+1}^{(1)} - d_t^{(1)} \quad (S1)$$

Basic operators. For a window x of length n ,

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i, \sigma(x) = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2}, R M S(x) = \sqrt{\frac{1}{n} \sum_{i=1}^n x_i^2}, E(x) = \sum_{i=1}^n x_i^2 \quad (S2)$$

$$R(x) = \max x - \min x, I Q R(x) = Q_{0.75} - Q_{0.25}, \\ M A D(x) = \text{median}(|x - \text{median}(x)|), A U C(x) = \sum_{i=1}^{n-1} \frac{x_i + x_{i+1}}{2} \quad (S3)$$

The standard deviation uses the population form ($1/n$), except for the global voltage sequence, which follows the publisher's supplementary definition with the sample form ($1/(n-1)$); both are labelled in the feature list.

Shape. Bias-corrected Fisher–Pearson skewness and excess kurtosis are used, with m_k the k -th central moment:

$$m_k = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^k, g_1 = \frac{m_3}{m_2^{3/2}}, g_2 = \frac{m_4}{m_2^2} - 3, \\ G_1 = \frac{\sqrt{n(n-1)}}{n-2} g_1, G_2 = \frac{n-1}{(n-2)(n-3)} [(n+1)g_2 + 6] \quad (S4)$$

Histogram entropy. The range of the window is divided into 16 equal bins and p_i is the fraction of samples in bin i :

$$H(x) = -\frac{1}{\ln 16} \sum_{p_i > 0} p_i \ln p_i \quad (S5)$$

Linear trend. A least-squares line is fitted against the sample index t_i :

$$a = \frac{\sum_i (x_i - \bar{x})(t_i - \bar{t})}{\sum_i (t_i - \bar{t})^2}, b = \bar{x} - a\bar{t}, R^2 = 1 - \frac{\sum_i (x_i - b - at_i)^2}{\sum_i (x_i - \bar{x})^2} \quad (S6)$$

Zero crossings and total variation. Zero crossings of a difference sequence are counted after removing exact zeros, giving y :

$$Z(y) = \#\{i: y_i y_{i+1} < 0\}, T V(x) = \sum_{i=1}^{n-1} |x_{i+1} - x_i| \quad (S7)$$

The 15 operators of Eqs. (S2)–(S7) applied to $S^{(1)}$, $S^{(2)}$ and $S^{(3)}$ give the 45 segment statistics; applied to the full sequence and to $d^{(1)}$ and $d^{(2)}$, together with end points and quantiles, they give the 24 global and 22 difference-dynamics features.

Spectral features. The demeaned sequence is Hann-windowed and transformed by a real fast Fourier transform ($N=101$, frequency in cycles per sample):

$$w_t = 0.5 - 0.5 \cos\left(\frac{2\pi t}{N-1}\right), x_t = (V_t - \bar{V}) w_t, \quad (S8)$$

$$X_k = \sum_{t=0}^{N-1} x_t e^{-j2\pi kt/N}, P_k = |X_k|^2, f_k = \frac{k}{N}$$

The amplitudes of the first eight non-DC lines and the energy shares of four bands are

$$A_k = \frac{2|X_k|}{\sum_t w_t} (k=1, \dots, 8), \quad (S9)$$

$$B_1 = \frac{\sum_{k=1}^3 P_k}{\sum_{k=1}^{50} P_k}, B_2 = \frac{\sum_{k=4}^7 P_k}{\sum_{k=1}^{50} P_k}, B_3 = \frac{\sum_{k=8}^{15} P_k}{\sum_{k=1}^{50} P_k}, B_4 = \frac{\sum_{k=16}^{50} P_k}{\sum_{k=1}^{50} P_k}$$

Spectral-shape features exclude the DC component: dominant frequency and amplitude, centroid, spread, flatness, entropy and 85% roll-off,

$$k^* = \arg \max_{k \geq 1} P_k, f_{dom} = f_{k^*}, A_{dom} = A_{k^*}, \quad (S10)$$

$$C = \frac{\sum_k f_k P_k}{\sum_k P_k}, S = \sqrt{\frac{\sum_k P_k (f_k - C)^2}{\sum_k P_k}}, F = \frac{\left(\prod_k P_k\right)^{1/K}}{\frac{1}{K} \sum_k P_k}$$

$$H_f = -\frac{1}{\ln 50} \sum_k p_k \ln p_k, p_k = \frac{P_k}{\sum_{\kappa} P_{\kappa}}, f_{85} = \min \left\{ f_k : \sum_{\kappa \leq k} P_{\kappa} \geq 0.85 \sum_{\kappa} P_{\kappa} \right\} \quad (S11)$$

Wavelet features. A four-level db2 decomposition with symmetric extension gives coefficient sets c , and their energy fractions are

$$\varepsilon_c = \sum_i c_i^2, EF_c = \frac{\varepsilon_c}{\sum_{c'} \varepsilon_{c'}}, c \in \{c A_4, c D_4, c D_3, c D_2, c D_1\} \quad (S12)$$

Eqs. (S8)–(S12) yield 24 frequency and wavelet features (8 amplitudes, 4 band shares, 7 spectral-shape features and 5 wavelet energy fractions). The dominant frequency takes a single value in the whole dataset and the roll-off frequency takes two values and is constant in the training cells of fold 0; their rank correlations are undefined in the affected folds and they are skipped by the finiteness condition of Section 2.3.3, which is why the feature-count ladder ends at 144 rather than 146.

Nonlinear features. Lagged autocorrelation ($\ell \in \{1, 5, 10\}$) and the three Hjorth parameters are

$$a = (V_0, \dots, V_{100-\ell}), b = (V_{\ell}, \dots, V_{100}), \rho_{\ell} = \frac{\sum_i (a_i - \bar{a})(b_i - \bar{b})}{\sqrt{\sum_i (a_i - \bar{a})^2 \sum_i (b_i - \bar{b})^2}}, \quad (S13)$$

$$Act = \sigma^2(V), Mob(V) = \sqrt{\frac{\sigma^2(d^{(1)})}{\sigma^2(V)}}, Com = \frac{Mob(d^{(1)})}{Mob(V)}$$

Approximate and sample entropy use embedding dimension $m=2$ (and $m+1=3$), tolerance $r=0.2\sigma(V)$ and the Chebyshev distance; approximate entropy counts self-matches, whereas sample entropy counts only pairs $i < j$:

$$C_i^m(r) = \frac{\#\{j: \max_{0 \leq k < m} |V_{i+k} - V_{j+k}| \leq r\}}{N - m + 1}, \Phi^m(r) = \frac{1}{N - m + 1} \sum_i \ln C_i^m(r),$$

$$A p E n = \Phi^2(r) - \Phi^3(r), S a m p E n = -\ln \frac{A}{B}$$
(S14)

Permutation entropy uses order 3 and delay 1, and the Hurst exponent is the slope of a variogram regression over lags $\ell \in \{1, 2, 4, 8, 16\}$:

$$P E = -\frac{1}{\ln 6} \sum_{\pi \in \Pi_3} p(\pi) \ln p(\pi), \tau(\ell) = \sqrt{\frac{1}{101 - \ell} \sum_i (V_{i+\ell} - V_i)^2},$$

$$H = \text{OLS slope of } \ln \tau(\ell) \text{ on } \ln \ell$$
(S15)

Transient-boundary features. Fifteen voltage differences between specified samples, including the steps at current changes ($V_1 - V_0$, $V_{31} - V_{30}$, $V_{41} - V_{40}$) and the net changes within segments ($V_{30} - V_1$, $V_{40} - V_{31}$, $V_{100} - V_{41}$), describe the transitions between segments. Three response times are defined on the rest segment $S^{[3]}$, with linear interpolation between samples and $\alpha \in \{0.10, 0.50, 0.90\}$:

$$p_t = \text{sgn}(V_{100} - V_{41})(V_t - V_{41}), t = 41, \dots, 100, t_\alpha = \min\{t: p_t \geq \alpha |V_{100} - V_{41}|\}$$
(S16)

Supplementary Note S2. PatchTST encoder equations

Each patch p_m of Eq. (12) is embedded by a shared linear map and a learnable positional embedding e_m , initialised from $N(0, 0.02^2)$ and trained with the network:

$$z_m^{(0)} = W_p p_m + b_p + e_m, W_p \in R^{D \times P}, Z^{(0)} = [z_1^{(0)}, \dots, z_M^{(0)}]^T \in R^{M \times D}$$
(S17)

The M tokens enter one Transformer encoder layer whose multi-head self-attention ($h=4$ heads, $d_k = D/h=8$) acts between patches:

$$Q_i = Z W_i^Q, K_i = Z W_i^K, V_i = Z W_i^V,$$

$$\text{head}_i = \text{softmax}\left(\frac{Q_i K_i^T}{\sqrt{d_k}}\right) V_i, M H A(Z) = [\text{head}_1; \dots; \text{head}_h] W^O$$
(S18)

The layer uses post-normalisation residual connections and a GELU feed-forward sublayer of width $2D=64$ with dropout 0.2:

$$Z' = L N(Z^{(0)} + M H A(Z^{(0)})), Z^{(1)} = L N(Z' + F F N(Z')),$$

$$F F N(u) = W_2 G E L U(W_1 u + b_1) + b_2$$
(S19)

Layer normalisation, mean pooling over tokens and a linear projection give the 32-dimensional waveform embedding:

$$E_v(\tilde{v}) = D r o p \left(R e L U \left(W_o \cdot \frac{1}{M} \sum_{m=1}^M L N(z_m^{(1)}) + b_o \right) \right) \in R^{32}$$
(S20)

Supplementary Note S3. Hyperparameter grids of the conventional regressors

Hyperparameters were selected on the validation cells of each fold by cell-macro RMSE, using the same validation cells as neural early stopping, and the selected models were not refitted on training plus validation cells. Ridge: regularisation coefficient from 10^{-4} to 10^3 . Random forest: 300 trees, maximum feature fraction $\in \{0.33, 1.0\}$, minimum leaf size $\in \{1, 5, 20\}$, seeds 0, 1, 2. GBDT (scikit-learn histogram gradient boosting): learning rate $\in \{0.05, 0.1\}$, maximum leaf nodes $\in \{15, 31\}$, minimum

leaf size $\in \{20, 100\}$, iterations $\in \{300, 1,000\}$. RBF-SVR: $C \in \{1, 10, 100\}$, insensitive-band width $\epsilon \in \{0.05, 0.1\}$ on the standardised target, kernel width by the scikit-learn “scale” rule. GPR: product of a constant and an RBF kernel plus a white-noise kernel; kernel hyperparameters were optimised by maximum likelihood on a fixed subsample of 2,000 training records (two random restarts), after which the model was conditioned on all training records. All algorithms except random forest are deterministic.

Supplementary Note S4. Statistical test families

Holm correction was applied within the following prespecified families at $\alpha = 0.05$: (1) the 14 encoder pairings versus MLP + CNN; (2) the five waveform complements versus A2; (3) the confirmatory ablations R0 – A1, R0 – A2, R0 – A3 and A3 – A4; (4) the three patch configurations versus R0; (5) the three adaptation ablations versus R0; (6) the six feature-count levels versus $K = 50$, separately for RMSE and MAE; (7) the conventional regressors, information-source controls, reproduced reference model and post hoc configurations versus A2 and versus R0 (30 comparisons); (8) the four residual comparisons on the GBDT backbone; and (9) the two post hoc raw-waveform augmentations. A1 – A2 was a single prespecified comparison without correction. Marginal effects, difference-in-differences, stratified increments, residual complementarity and low-SOH bias are descriptive. Confidence intervals are conditional on the fixed split and the trained models and do not cover the variability of new data collection or repartitioning.

Supplementary Note S5. Waveform increment over the GBDT backbone

With the GBDT selected in Supplementary Note S3 as the backbone, networks with the structures of R0, A2 and A1 learned its residuals (main text, Section 3.5.4). The waveform increment (G+R0) – (G+A2) was -0.008 pp with 36 of 64 cells improved (Holm $p = 0.35$), and none of the residual learners improved significantly on the GBDT alone. Appending the 101 standardised voltages directly as additional input columns of GBDT (extended grid with up to 63 leaf nodes and 3,000 iterations) and of RBF-SVR, followed by re-tuning, likewise produced no detectable gain. These analyses were defined after part of the GBDT results had been seen and are reported as post hoc.

Supplementary Note S6. Re-implementation used for the generalisation, robustness and deployment experiments

The experiments of Section 4.6 were run with an independent re-implementation of the protocol written in Python (NumPy, scikit-learn 1.9 and PyTorch 2.14 on CPU, one thread per run), starting from the publisher’s processed pulse files [31]. Cleaning reproduced the record counts exactly: 17,130 records in 2,855 groups before cleaning, 192 records removed at the 42nd RPT, and 16,938 records in 2,823 complete groups (SOH 72.34–100%) after cleaning. The 143 waveform features were implemented from the definitions in Supplementary Note S1; details not fixed by those definitions, such as the quantile levels of the global family and the nine voltage differences not listed in Note S1, may differ from the original code, and the response-time levels were set to 0.5, 0.632 and 0.9. Feature selection was nonetheless similar: in fold 0 it selected 16 frequency and wavelet, 14 segment, 11 difference-dynamics, 6 global, 1 boundary and 1 nonlinear feature together with the DCIR, which ranked 13th; 44 features were selected in every fold and 57 in at least one. The split permutation used NumPy’s default generator with seed 42, which reproduces the fold sizes but not necessarily the fold membership of the original runs. Architectures, loss, optimiser, scheduler, early stopping and seeds follow Sections 3.2–3.4 of the main text, with positional embeddings initialised from $U(-0.02, 0.02)$.

Under the original five-fold protocol the re-implementation gives R0 1.427, A2 1.474 and GBDT 1.028 pp (reported: 1.377, 1.500 and 1.022 pp). The waveform increment keeps its sign and significance

(-0.046 pp, -0.070 to -0.023 ; $p < 0.001$; 42 of 64 cells), and cells 12 and 3 are again the best- and worst-estimated cells. Table S7 lists all re-implementation results. Among the conventional baselines, ridge regression, RBF-SVR, the τ -only controls and the without- τ ridge and GBDT were re-run (Table S5); where reported values exist, they agree closely (RBF-SVR 1.007 vs 1.026 pp, without- τ GBDT 1.698 vs 1.663 pp, τ -only models about 3 pp). Random forest and GPR on the full input and the without- τ random forest were not re-run because of their computational and memory cost.

NMC/graphite dataset. The UConn-ILCC NMC dataset [31] records the same 100-s pulses at nine nominal SOCs; the three SOCs used for LFP (20%, 50% and 90%) were retained, giving six conditions per RPT, 8,016 records in 1,336 cell-RPT groups of 44 cells and SOH from 41.2% to 100%. Because NMC cells have a higher resistance, the voltage-step and range limits of Eq. (2) were relaxed to 0.3 and 0.8 V; no record was removed. The pipeline was otherwise unchanged. On these data the two neural models perform alike (R0 2.030, A2 2.026 pp; R0 – A2 = +0.004 pp, -0.042 to 0.047 ; $p = 0.90$), both clearly better than A1 (2.317 pp), and GBDT reaches 0.935 pp; GBDT is more accurate than the neural models in every SOH range.

Noise not seen in training. When Gaussian noise is added only to the test voltages of models trained on clean data, all feature-based models deteriorate strongly ($\sigma_{\text{noise}} = 0.5$ mV: R0 4.37, A2 4.60, GBDT 1.48 pp), because difference- and spectrum-based features computed on the smooth, 0.1-mV-resolution resampled records are sensitive to noise absent from training. Models should therefore be trained on data from the target sensor, or with matching noise augmentation, as in the matched-noise experiment of Section 4.6.

Supplementary Algorithm S1

Algorithm S1. Training and inference of the dual-branch SOH estimation model (handcrafted backbone + pulse-waveform complement).

Input: cleaned record set $D = \{(V, x, y, \text{cell}, \text{rpt}, \text{soc}, \text{pulse})\}$ with 16,938 records of 64 cells; $F = 5$ folds, split seed 42, initialisation seeds $\{0, 1, 2\}$; $K = 50$, redundancy threshold 0.98, $\lambda = 0.1$.

Output: record-level test predictions and cell-macro errors for every fold and seed.

Offline training

- 1: Remove cumulative cycle count, RPT index, coulomb-counted SOC and SOC deviation from the 150 candidates $\rightarrow$ 146 selectable features (Section 2.3.2)
- 2: Permute the 64 cells with seed 42 and divide them into five disjoint test folds (13/13/13/13/12 cells)
- 3: In each fold, draw validation cells from the non-test cells with the fold index as offset and stride 5 (train/validation/test = 40/11/13, 41/10/13, 41/10/13, 41/10/13, 42/10/12)
- 4: **for** outer fold $f = 0, \dots, 4$ **do**
- 5: Split D by cell into D_{tr}, D_{va}, D_{te} ; all RPTs and condition records of a cell stay together
- 6: Compute Spearman ρ_j of the 146 candidates with SOH on D_{tr} only $\triangleright$ Eq. (4)
- 7: Select K features greedily by $|\rho_j|$, skipping rank correlation ≥ 0.98 with selected ones; fill to K if needed $\triangleright$ Eq. (5)
- 8: Fit medians, feature means/s.d., per-position waveform means/s.d., cycle-count and label statistics on D_{tr} only
- 9: Transform D_{tr}, D_{va}, D_{te} with these statistics $\triangleright$ Eqs. (6)–(10)
- 10: **for** seed $s \in \{0, 1, 2\}$ **do**
- 11: Initialise E_h, E_v and g with seed s ; AdamW (lr 10^{-3} , weight decay 10^{-4}), batch 256, ReduceLROnPlateau (patience 5, factor 0.5), gradient clipping 5, at most 100 epochs, early-stopping patience 15
- 12: **for** each epoch and mini-batch **do**
- 13: Enable gradients on τ ; forward pass gives $\hat{y}^*$ $\triangleright$ Eq. (11)
- 14: Compute $\partial \hat{y}^* / \partial \tau$ by automatic differentiation (create_graph = True)
- 15: $\mathcal{L} \leftarrow \text{MSE}(\hat{y}^*, y^*) + \lambda \cdot \{[\partial \hat{y}^* / \partial \tau]_+^2\}$; back-propagate, clip, update $\triangleright$ Eq. (13)
- 16: After each epoch, compute the cell-macro RMSE on D_{va} ; keep the best weights; stop after 15 epochs without improvement
- 17: Load the best-epoch weights and predict D_{te}

Online inference (one pulse record)

- 18: Reference the 101-point voltage to its first sample and standardise with the fold statistics $\rightarrow \tilde{v} \triangleright$ Eqs. (6), (7)
- 19: Compute the K selected features, impute and standardise, append standardised $\tau \rightarrow \mathbf{h} \triangleright$ Eqs. (8), (9)
- 20: $\hat{y}^* \leftarrow g([E_v(\tilde{v}); E_h(\mathbf{h})])$; SOH $\leftarrow 100(\sigma_{\hat{y}} \hat{y}^* + \mu_y) \triangleright$ Eqs. (11), (10)

Metrics and statistics

- 21: Average the six condition predictions of each cell–RPT, compute per-cell RMSE and the cell-macro mean $\triangleright$ Eq. (14)
- 22: Paired differences: 5,000 fold-stratified cell bootstrap intervals; 5,000 two-sided paired sign-flip tests; Holm correction within families

Supplementary Tables

Supplementary Table S1. Ageing protocols of the cycling-condition groups.

Group	CC–CV charge rate (C)	CC discharge rate (C)	DOD (%)	Cells (IDs)
1	2	1.5	60	6 (2–7)
2	2.42	1.5	60	6 (8–13)
3	2	2	60	6 (14–19)
4	2	1.5	80	6 (20–25)
5	2.83	1.5	60	6 (26–31)
6	2.42	2	60	6 (32–37)
7	2.42	1.5	80	6 (38–43)
8	3.25	1.5	60	6 (44–49)
9	2	1.5	100	6 (50–55)
10	2	2	80	6 (56–61)
11	2.83	1.5	80	4 (62–65)

From the cell test log in the publisher’s public repository [28]. Charging is CC–CV and discharging CC; rates refer to the nominal 1.2 Ah, so 2.42C, 2.83C and 3.25C correspond to 2.9 A, 3.4 A and 3.9 A.

Supplementary Table S2. Input information of the two branches.

Input	Dimension	Waveform branch	Backbone branch	Source
First-sample-referenced voltage sequence	101	✓	—	Single pulse
Waveform-derived features (train-fold selected)	49	—	✓	Single pulse
Matched-condition DCIR (selected)	1	—	✓	Single pulse
Standardised cumulative cycle count τ	1	—	✓	Cycle record
Nominal SOC, pulse direction	—	—	Candidate, not selected	Test setting
Coulomb-counted SOC, SOC deviation	—	—	Excluded	Requires reference capacity
Other condition records, past capacities, cell identity	—	—	—	—

✓, input enters the branch; —, not used or not applicable.

Supplementary Table S3. Ablation configurations and reference models.

Category	Configuration	Handcrafted backbone	Waveform complement	τ	λ	Trainings
Main	R0	MLP, 50 features	PatchTST (9, 4)	✓	0.1	15
Encoder matrix	15 pairings	MLP / ResNet / FT-Transformer	CNN / MLP / TCN / PW-Trans. / PatchTST	✓	0.1	225 (incl. R0)
Branch ablation	A2	MLP	—	✓	0.1	15
Branch ablation	A1	DCIR only (concatenated)	PatchTST (9, 4)	✓	0.1	15
Factor ablation	A3	MLP	PatchTST (9, 4)	✓	0	15
Factor ablation	A4	MLP	PatchTST (9, 4)	—	0	15
Patch parameters	P1, P2, P3	MLP	PatchTST (5, 2), (5, 4), (9, 2)	✓	0.1	45
Adaptation	D1, D2, D3	MLP	Flatten head, + RevIN, raw voltage	✓	0.1	45
Feature count	$K = 10, 20, 30, 75, 100, 144$	MLP, K features	PatchTST (9, 4)	✓	0.1	90 ($K = 50$ reuses R0)
Conventional	Ridge, RF, GBDT, SVR, GPR	Same 51-dim input as A2	—	✓	—	1 per fold (RF 3)
Information source	τ only	τ only	—	✓	—	1 per fold (RF 3)

Category	Configuration	Handcrafted backbone	Waveform complement	τ	λ	Trainings
Information source	Without τ	50 features	—	—	—	1 per fold (RF 3)
Published model	Nowacki et al. [28], reproduced	—	Raw ΔV , 5×100 MLP	—	—	30 models
Strong backbone	G+R0, G+A2, G+A1	GBDT, network learns residual	As R0, A2, A1	✓	0.1	45

Neural configurations use 5 folds $\times$ 3 seeds (510 trainings in total, plus 30 reproduced models). Post hoc GBDT/SVR raw-voltage configurations are not listed.

Supplementary Table S4. Stratified waveform increment (R0 – A2, SOH pp).

Stratum	Level	A2 RMSE	R0 RMSE	Δ RMSE	95% CI	Relative change	Improved cells
True SOH	<80%	2.404	1.991	-0.413	-0.460 to -0.363	-17.2%	62/64
True SOH	80–85%	1.742	1.601	-0.141	-0.200 to -0.082	-8.1%	47/64
True SOH	85–90%	1.974	1.861	-0.113	-0.151 to -0.074	-5.7%	50/64
True SOH	90–95%	1.150	1.123	-0.027	-0.046 to -0.009	-2.3%	36/64
True SOH	$\geq 95\%$	0.956	0.808	-0.148	-0.163 to -0.134	-15.5%	61/64
Nominal SOC	20%	1.546	1.452	-0.094	-0.116 to -0.072	-6.1%	55/64
Nominal SOC	50%	1.550	1.414	-0.136	-0.161 to -0.111	-8.8%	58/64
Nominal SOC	90%	1.579	1.474	-0.104	-0.127 to -0.080	-6.6%	56/64
Direction	Charge	1.546	1.434	-0.112	-0.136 to -0.088	-7.2%	55/64
Direction	Discharge	1.584	1.477	-0.107	-0.125 to -0.088	-6.7%	57/64
Condition	20% charge	1.533	1.431	-0.102	-0.128 to -0.076	-6.7%	47/64
Condition	20% discharge	1.556	1.471	-0.085	-0.109 to -0.061	-5.5%	49/64
Condition	50% charge	1.512	1.380	-0.132	-0.161 to -0.104	-8.7%	57/64
Condition	50% discharge	1.578	1.439	-0.139	-0.167 to -0.110	-8.8%	54/64
Condition	90% charge	1.567	1.458	-0.109	-0.141 to -0.078	-7.0%	52/64
Condition	90% discharge	1.584	1.483	-0.101	-0.124 to -0.077	-6.4%	54/64

True-SOH strata are formed at the cell–RPT level after averaging the six conditions, and each cell’s RMSE is computed within the stratum; SOC, direction and condition strata use single records. Predictions are averaged over the three seeds. Descriptive analysis without multiplicity correction.

Supplementary Table S5. Conventional regressors, information-source controls and reference models (SOH pp).

Model	Input	Reported	Re-implemented
R0 (dual-branch)	Features + τ + ΔV	1.377	1.427
A2 (backbone only)	Features + τ	1.500	1.474
Ridge	Features + τ	n.r.	1.879
Random forest	Features + τ	n.r.	—
GBDT	Features + τ	1.022	1.028
RBF-SVR	Features + τ	1.026	1.007
GPR	Features + τ	n.r.	—
Ridge	τ only	$\approx 3^a$	3.083
Random forest	τ only	$\approx 3^a$	3.095
GBDT	τ only	$\approx 3^a$	2.881
Ridge	Features without τ	n.r.	2.593
Random forest	Features without τ	n.r.	—
GBDT	Features without τ	1.663	1.698
Nowacki et al. [28], reproduced	Raw ΔV	1.628	—
GBDT + raw voltage (post hoc)	Features + τ + 101 voltages	no gain ^b	—
RBF-SVR + raw voltage (post hoc)	Features + τ + 101 voltages	no gain ^b	—

Cell-macro RMSE over the 64 test cells. “Reported” values are those of the original runs on which the main text is based; n.r., not given in the source draft (to be inserted from the original result files). “Re-implemented” values come from the CPU re-implementation of Supplementary Note S6 with the same grids (random forest averaged over seeds 0–2); —, not run in the re-implementation. ^a About 3 pp in the original runs. ^b No detectable gain over GBDT and RBF-SVR, respectively (Supplementary Note S5).

Supplementary Table S6. Sensitivity to the number of handcrafted features (SOH pp).

K	Backbone input	Parameters	RMSE	MAE	Δ RMSE	95% CI	Holm p	Improved / worse cells
10	11	16,481	1.516	1.249	+0.139	0.089–0.188	0.001	16/48
20	21	17,121	1.499	1.226	+0.122	0.077–0.169	0.001	14/50
30	31	17,761	1.468	1.210	+0.091	0.057–0.125	0.001	14/50
50 (R0)	51	19,041	1.377	1.141	—	—	—	—
75	76	20,641	1.414	1.165	+0.037	0.018–0.057	0.002	24/40
100	101	22,241	1.418	1.176	+0.041	0.021–0.062	0.002	15/49
144	145	25,057	1.376	1.128	−0.002	−0.019 to 0.015	0.838	32/32

K excludes the separately appended cycle count, so the backbone input has $K + 1$ dimensions. Δ RMSE is the level minus $K = 50$; Holm correction covers the six RMSE comparisons. Improved (worse) cells have lower (higher) error than $K = 50$.

Supplementary Table S7. Results of the re-implementation (cell-macro RMSE, SOH pp).

Scenario	Data	R0	A2	A1	GBDT	R0 – A2	95% CI	p	Improved cells
Five-fold, reported (main text)	LFP	1.377	1.500	1.494	1.022	−0.123	−0.141 to −0.103	<0.001	58/64
Five-fold, re-implementation	LFP	1.427	1.474	—	1.028	−0.046	−0.070 to −0.023	<0.001	42/64
Leave-one-group-out	LFP	1.560	1.594	—	1.401	−0.035	−0.053 to −0.015	0.008	39/64
Cycle count +10% (test)	LFP	1.449	1.497	—	1.117	−0.048	−0.072 to −0.024	<0.001	43/64
Cycle count −10% (test)	LFP	1.477	1.518	—	1.176	−0.041	−0.063 to −0.018	0.002	39/64
Matched noise, $\sigma_{\text{noise}} = 1$ mV	LFP	1.620	1.678	—	1.372	−0.058	−0.077 to −0.040	<0.001	50/64
Matched noise, $\sigma_{\text{noise}} = 2$ mV	LFP	1.715	1.760	—	1.424	−0.045	−0.069 to −0.022	0.002	46/64
Test-only noise, $\sigma_{\text{noise}} = 0.5$ mV ^a	LFP	4.374	4.595	—	1.485	—	—	—	—
Five-fold cell-disjoint	NMC	2.030	2.026	2.317	0.935	+0.004	−0.042 to 0.047	0.90	21/44

Paired comparisons use 5,000 fold- (or group-) stratified cell bootstraps and two-sided sign-flip tests, without multiplicity correction. ^a Models trained on clean data with noise added to the test voltages only. Group-level leave-one-group-out errors are shown in Fig. 8a of the main text.

Supplementary Figures

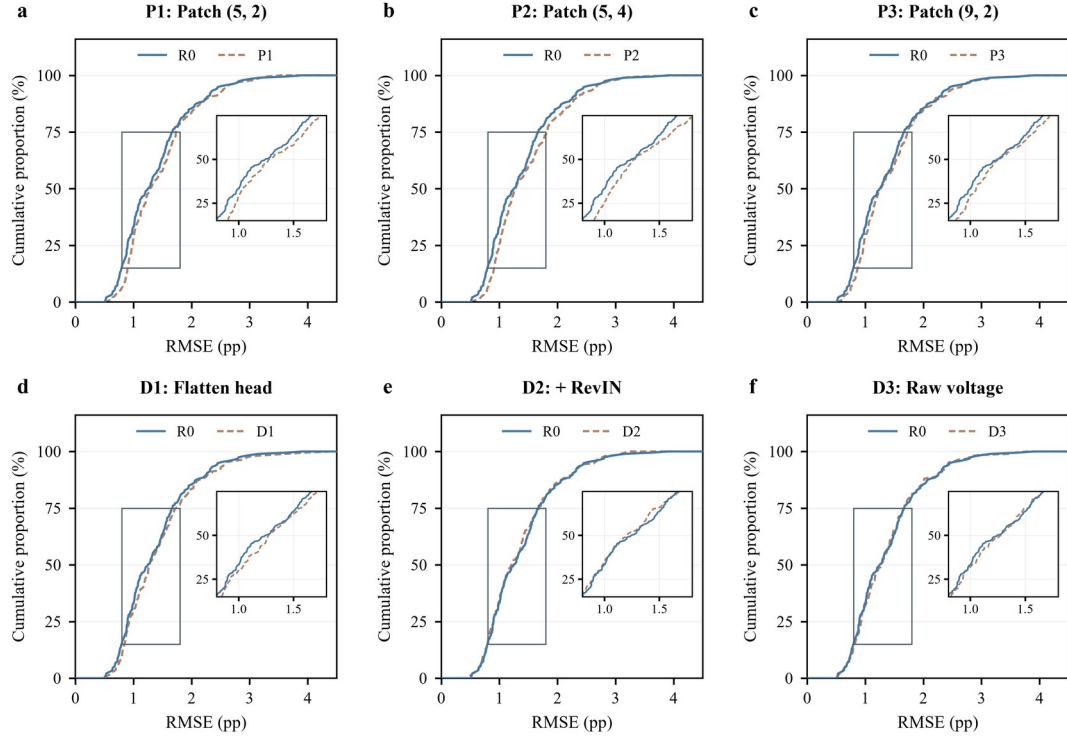


Fig. S1. Cumulative error distributions of the waveform-branch design variants. (a–c) Patch configurations P1 (5, 2), P2 (5, 4) and P3 (9, 2) versus R0 (9, 4); (d–f) adaptation ablations D1 (flatten head), D2 (with RevIN) and D3 (raw voltage). Solid blue, R0; dashed brown, variant. Each curve contains the 192 cell $\times$ seed RMSEs (not 192 independent cells); insets magnify RMSE 0.8–1.8 pp and cumulative proportion 15–75%. Cell-level tests are given in Table 8.

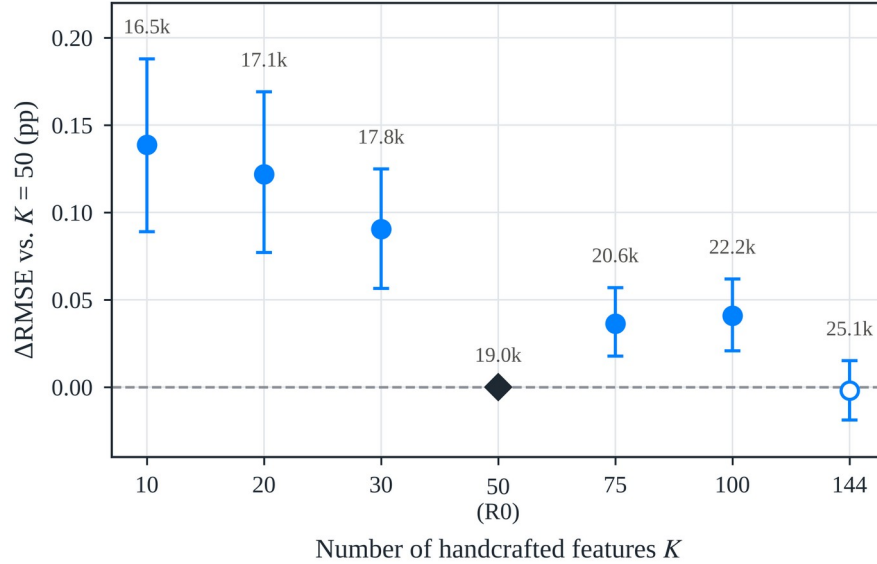


Fig. S2. Sensitivity to the number of handcrafted features K . Difference in cell-macro RMSE relative to $K = 50$ with 95% fold-stratified cell bootstrap intervals; labels give the total parameter count. Hollow markers denote intervals that include zero.

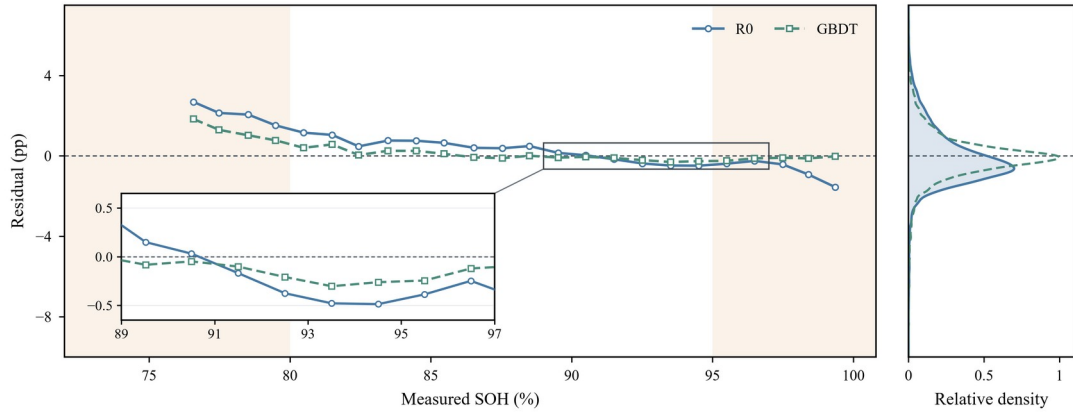


Fig. S3. SOH-dependent mean bias and overall residual distribution of R0 (blue, solid) and GBDT (teal, dashed). Residuals are the seed-averaged prediction minus the measured SOH of each record. Left: 1-pp bins of measured SOH with at least 20 records; shading marks SOH < 80% and ≥ 95%; the inset magnifies 89–97% SOH. Right: kernel density of all 16,938 residuals, normalised to the common maximum.

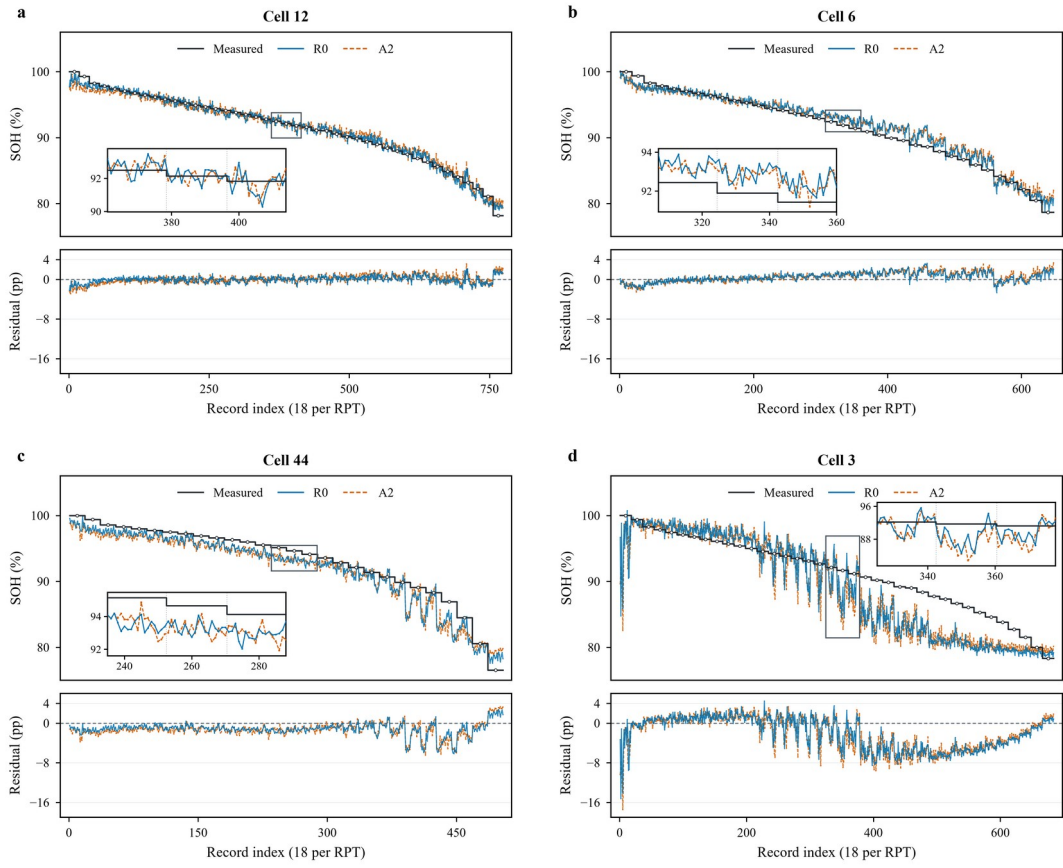


Fig. S4. Record-level predictions and residuals of representative cells 12, 6, 44 and 3 (ranks 1, 22, 43 and 64 by R0 error). Each RPT contributes 18 records (three SOCs × two directions × three seeds), so the horizontal axis is the record index rather than the cycle count; the grey staircase repeats the measured SOH of each RPT. Blue, R0; orange dashed, A2.

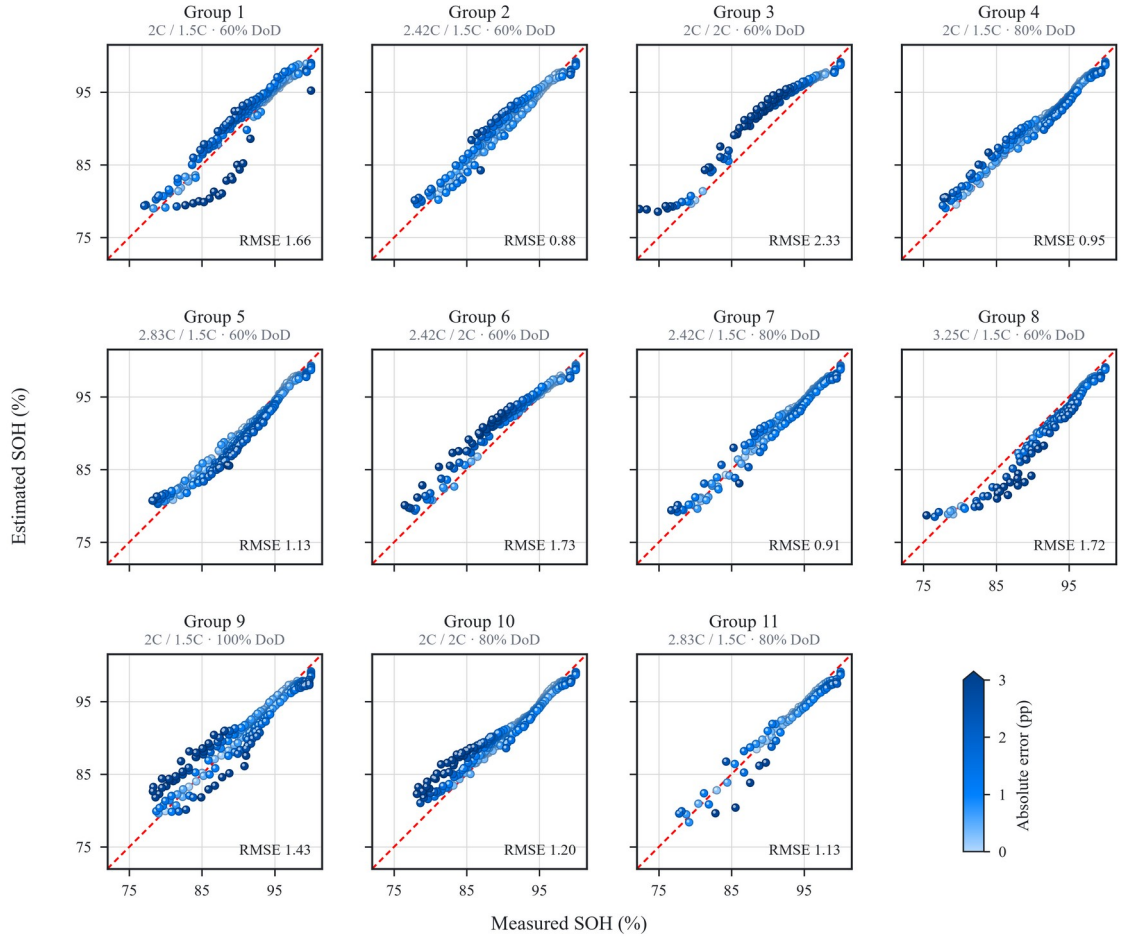


Fig. S5. SOH estimated by R0 versus measured SOH for the 11 cycling-condition groups. Each point is one cell-RPT (predictions averaged over seeds and conditions, all from the cell's test fold); colour denotes absolute error (saturating at 3 pp) and the red dashed line is $y = x$. The number in each panel is the group's cell-macro RMSE following Eq. (14).